

SHOPEZ – FULL STACK MERN E-COMMERCE PLATFORM

1. INTRODUCTION

Project Title : ShopEZ – Full Stack MERN E-Commerce Platform

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Role: Full Stack Developer

2. PROJECT OVERVIEW

Purpose

ShopEZ is a full-stack e-commerce web application developed using the MERN stack (MongoDB, Express.js, React.js, and Node.js). The main objective of this project is to provide users with a simple and secure online shopping platform where they can browse products, create an account, manage their cart, and place orders easily.

This project helped me understand how the frontend, backend, and database work together to build a complete web application.

Key Features

- User Registration and Login
- Secure JWT Authentication
- Browse Products by Category
- Search Products
- Shopping Cart Management
- Wishlist
- Checkout Process
- Order Management
- Responsive User Interface
- Admin Dashboard
- Product & Category Management
- MongoDB Database Integration

3. SYSTEM ARCHITECTURE

Frontend

The frontend is developed using **React.js** and **Vite**. It provides a clean and responsive user interface where users can browse products, manage their shopping cart, and place orders.

Backend

The backend is built using **Node.js** and **Express.js**. It handles all business logic, user authentication, product management, and communication with the database using REST APIs.

Database

MongoDB is used to store all application data, including users, products, orders, categories, cart details, and wishlist items.

Architecture Flow

React Frontend

↓

Express REST API

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Node.js Server

↓

MongoDB Database

4. SETUP INSTRUCTIONS

Software Requirements

- Node.js (v18 or above)
- npm
- MongoDB Community Server
- MongoDB Compass
- Visual Studio Code
- Git

5. PROJECT FOLDER STRUCTURE

Frontend

- assets
- components
- context
- layouts
- pages
- styles
- App.jsx
- main.jsx

Backend

- config
- controllers
- middleware
- models
- routes
- uploads
- utils
- server.js

6. RUNNING THE APPLICATION

Start Backend

```
cd backend  
npm run dev
```

Start Frontend

```
cd frontend  
npm run dev
```

Application URL

<http://localhost:5173>

7. API DOCUMENTATION

Authentication APIs

- POST /api/auth/register
- POST /api/auth/login
- GET /api/auth/me

Product APIs

- GET /api/products
- GET /api/products/:id

Cart APIs

- POST /api/cart
- GET /api/cart
- DELETE /api/cart/:id

Order APIs

- POST /api/orders
- GET /api/orders

Admin APIs

- GET /api/admin/products
- GET /api/admin/orders
- GET /api/admin/users

8. USER AUTHENTICATION

Authentication is implemented using **JSON Web Tokens (JWT)**.

Features

- User Registration
- User Login
- Password Encryption using bcrypt.js
- JWT Token Generation
- Protected Routes
- Role-Based Access Control (User/Admin)

9. USER INTERFACE

The application consists of the following pages:

- Home Page
- Product Categories
- Product Listing
- Product Details
- Shopping Cart
- Wishlist
- Login Page
- Registration Page
- Checkout Page
- User Profile
- Admin Dashboard

10. TESTING

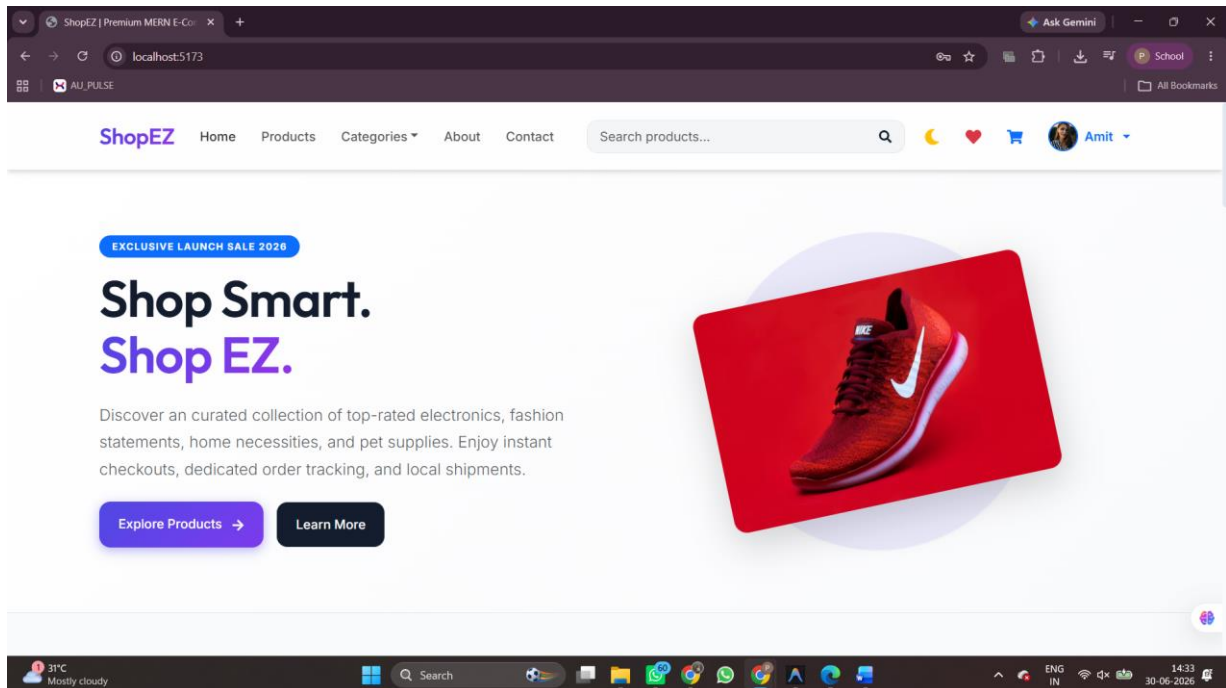
The application was tested to ensure all major functionalities work correctly.

Testing Performed

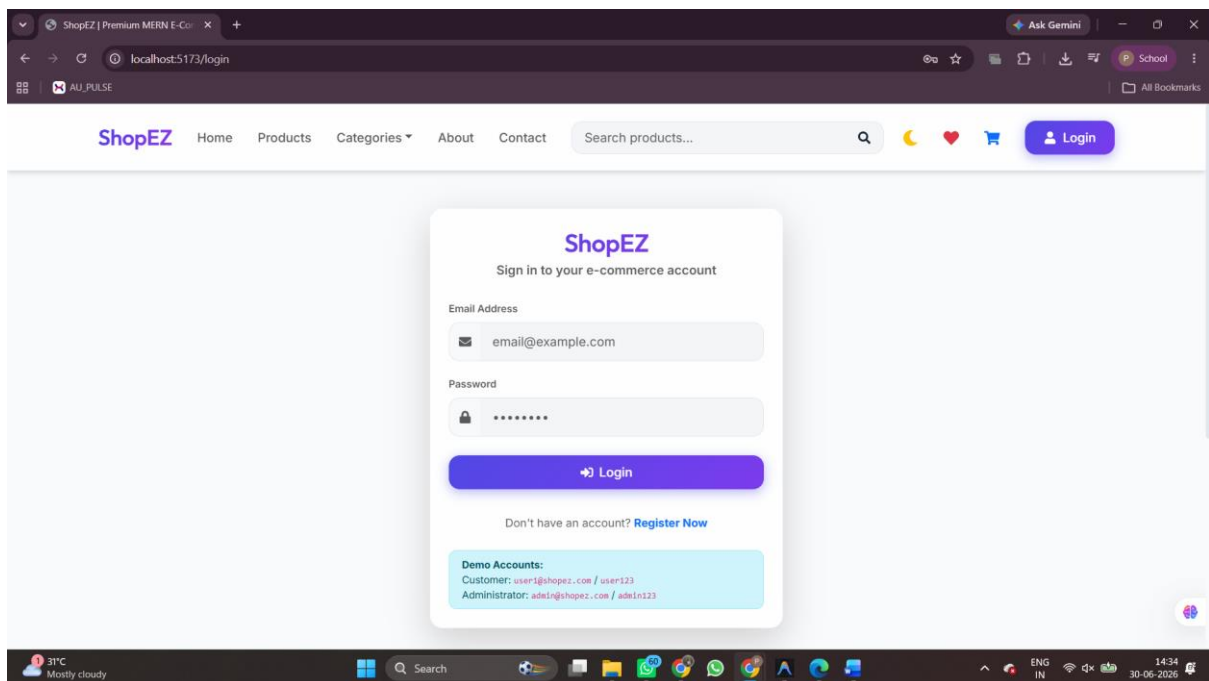
- User Registration
- User Login
- Product Search
- Product Filtering
- Add to Cart
- Remove from Cart
- Checkout Process
- Admin Dashboard
- Product Management
- API Testing

11. SCREENSHOTS

- Home Page



- Login Page



- Registration Page

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localhost:5173/register

ShopEZ Home Products Categories About Contact Search products...

Create your customer account

Full Name
Nikhil Ravva

Email Address
email@example.com

Password

Confirm Password

Register

Already have an account? [Sign In](#)

31°C Mostly cloudy

- Product Listing

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localhost:5173/products

ShopEZ Home Products Categories About Contact Search products...

Browse Products
Found 100 products matching your selections

Sort by: Newest Arrivals

Filter Options Clear

KEYWORD
Search...

CATEGORY
All Categories

BRAND
All Brands

PRICE RANGE
Min - Max

MINIMUM RATING
Any Rating

38% OFF
HP Comfort Heels
Footwear
₹531

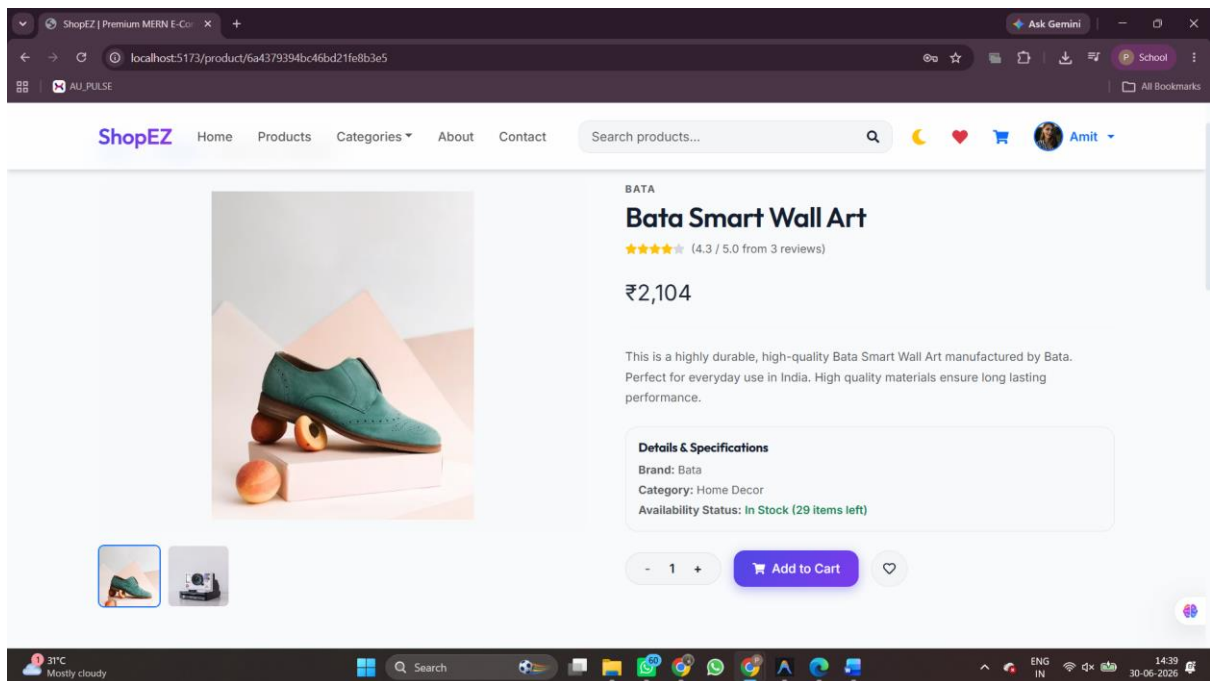
BOAT
Boat Elegant Notebook
Books
₹3,618

17% OFF
PIGEON
Pigeon Authentic Lamp
Home Decor
₹2,708

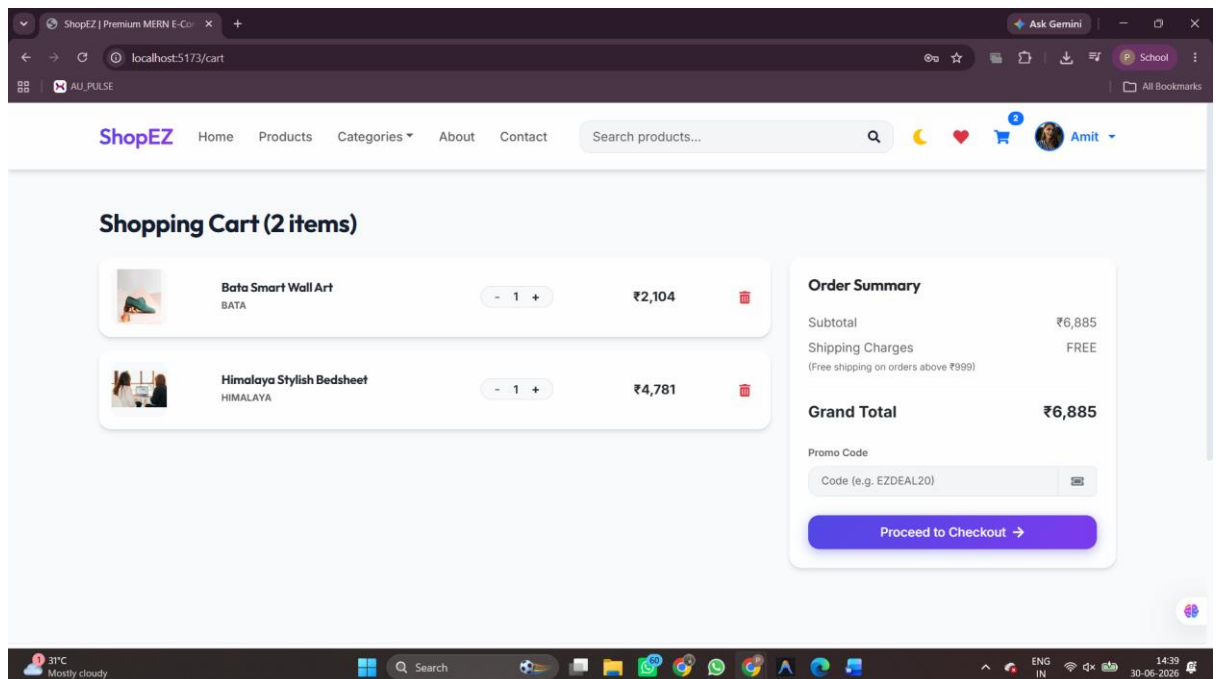
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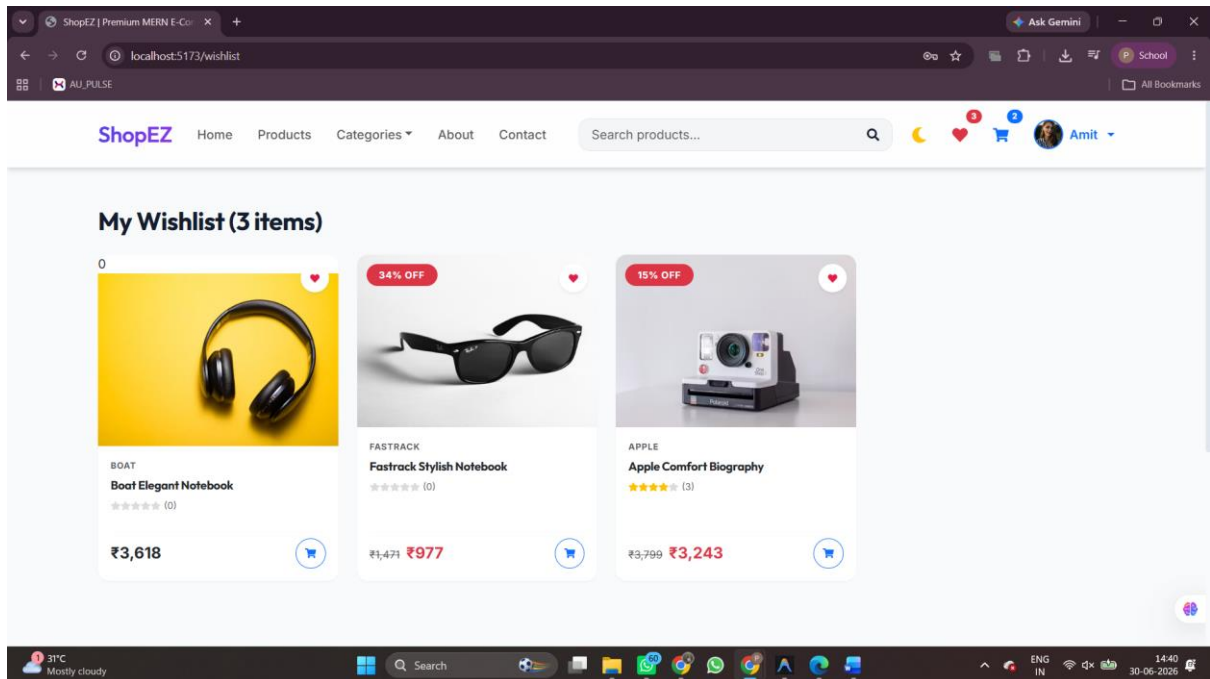
- Product Details



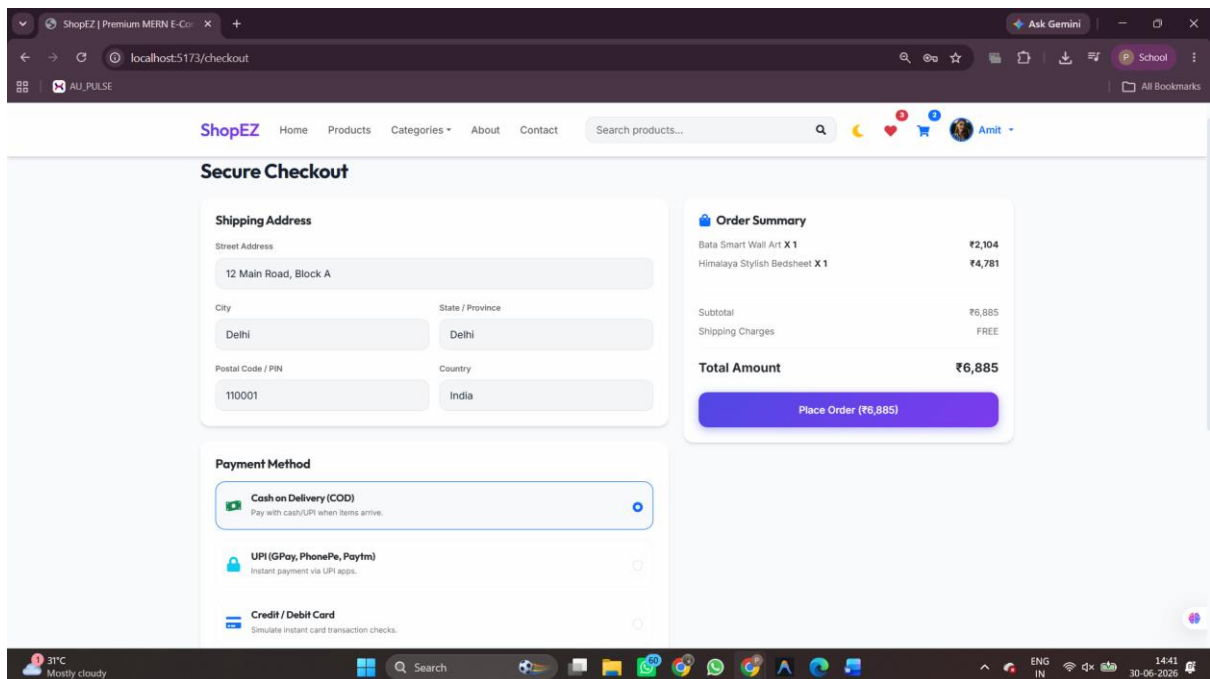
- Shopping Cart



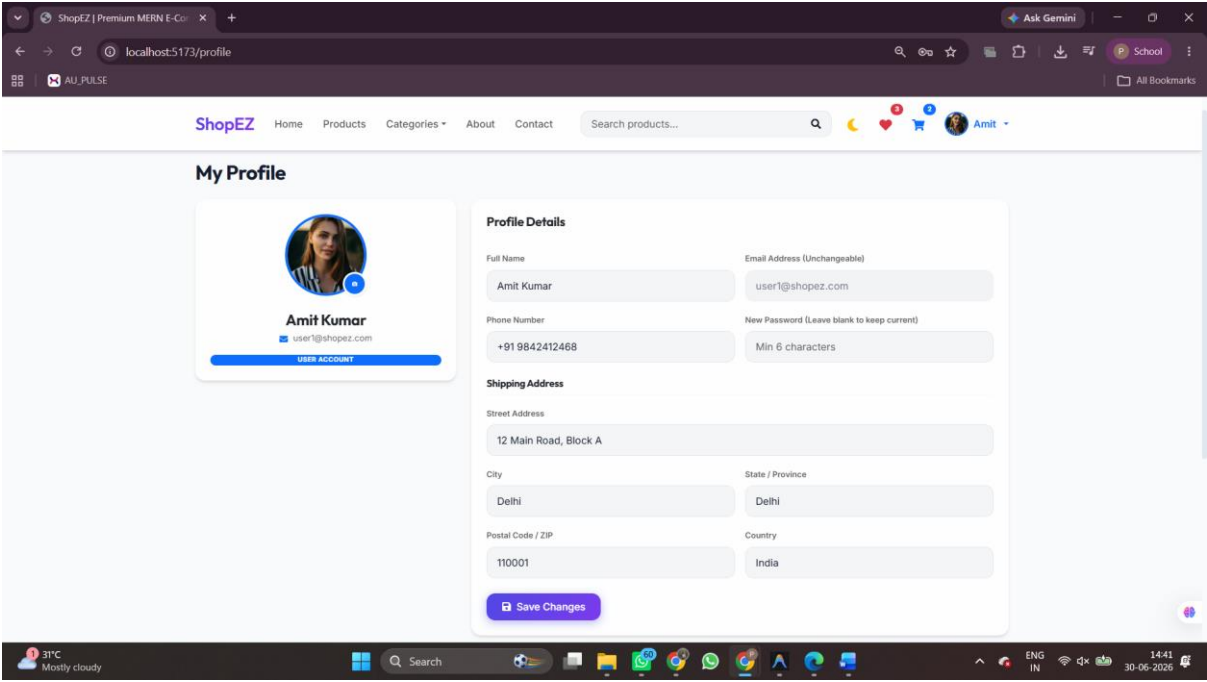
- Wishlist



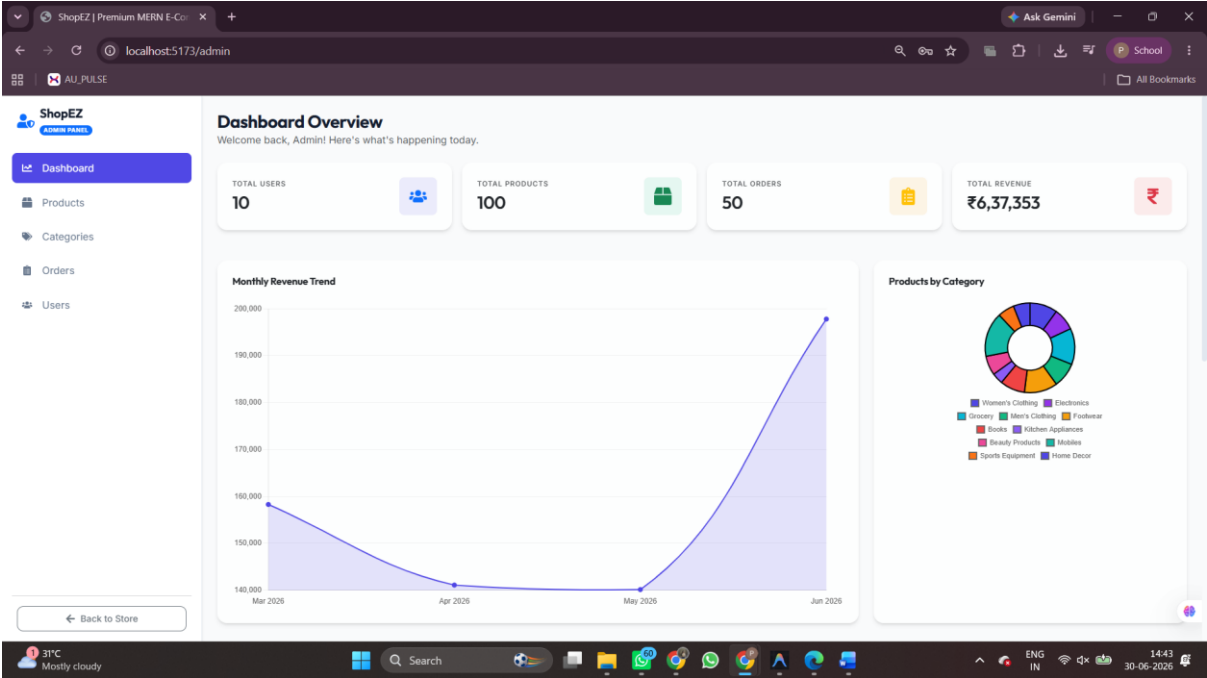
- Checkout Page



- User Profile



- Admin Dashboard



12. KNOWN ISSUES

- MongoDB must be running while using the project locally.
- Image loading may depend on internet connectivity.
- Online payment gateway is currently simulated.
- Some features require admin login.

13. FUTURE ENHANCEMENTS

The following features can be added in future versions:

- Razorpay Payment Integration
- Email Notifications
- Product Reviews & Ratings
- Live Order Tracking
- AI-based Product Recommendations
- Mobile Application
- Multi-language Support
- Advanced Sales Analytics
- Cloud Deployment using AWS or Azure